

13.6

Directional derivatives & gradients

Example 1

Let $f(x, y) = xy$ Find $D_v f(1, 2)$ for unit vector

$$v = \frac{\sqrt{3}}{2} i + \frac{1}{2} j$$

Sol:- $f(x, y) = xy$

$$\frac{\partial f}{\partial x} = f_x = y$$

$$f_x|_{(1, 2)} = 2$$

$$\frac{\partial f}{\partial y} = f_y = x$$

$$f_y|_{(1, 2)} = 1$$

$$D_u f = \Delta f \cdot \hat{u}$$

$$\Delta f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}$$

or

$$f_x \hat{i} + f_y \hat{j} + f_z \hat{k}$$

putting values

$$2\hat{i} + \hat{j}$$

$$\hat{u} = \frac{\vec{u}}{|\vec{u}|} = \frac{\frac{\sqrt{3}}{2} \hat{i} + \frac{1}{2} \hat{j}}{\sqrt{\left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{2}\right)^2}}$$

$$= \frac{\frac{\sqrt{3}}{2} \hat{i} + \frac{1}{2} \hat{j}}{\sqrt{1}}$$

$$= \frac{\frac{\sqrt{3}}{2} \hat{i} + \frac{1}{2} \hat{j}}{1}$$

$$= \frac{\sqrt{3}}{2} \hat{i} + \frac{1}{2} \hat{j}$$

$$D_v f = (2\hat{i} + \hat{j}) \cdot \left(\frac{\sqrt{3}}{2} + \frac{1}{2}\hat{j} \right)$$

$$= 2 \cdot \frac{\sqrt{3}}{2} (\hat{i} \cdot \hat{i}) + 1 \cdot \frac{1}{2} (\hat{j} \cdot \hat{j})$$

$$= \frac{\sqrt{3} + 1}{2}$$

Example 3

Find the directional derivative of $f(x, y, z) = x^2y - yz^3 + z$ at the point $(1, -2, 0)$ in the direction of the vector $a = 2\hat{i} + \hat{j} - 2\hat{k}$

Sol:-

$$\frac{\partial f}{\partial x} = f_x(x, y, z) = 2xy, \quad f_x|_{(1, -2, 0)} = -4$$

$$\frac{\partial f}{\partial y} = f_y = x^2 - z^3, \quad f_y|_{(1, -2, 0)} = 1$$

$$\frac{\partial f}{\partial z} = f_z = -3yz^2 + 1, \quad f_z|_{(1, -2, 0)} = 1$$

$$D_{\vec{a}} f = \Delta f \cdot \hat{u}$$

$$\Delta f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}$$

$$= -4\hat{i} + \hat{j} + \hat{k}$$

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|}$$

$$\hat{a} = \frac{2\hat{i} + \hat{j} - 2\hat{k}}{\sqrt{(2)^2 + (1)^2 + (-2)^2}} = \frac{2\hat{i} + \hat{j} - 2\hat{k}}{\sqrt{9}}$$

$$D_{\hat{a}} f = \left(-4\hat{i} + \hat{j} + \hat{k} \right) \cdot \left(\frac{2\hat{i} + \hat{j} - 2\hat{k}}{3} \right)$$

$$= -4\left(\frac{2}{3}\right) + \frac{1}{3} - \frac{2}{3} = -3 \text{ Ans}$$

Example 4

Let $f(x, y) = x^2 e^y$. Find the maximum value of a directional derivative at $(-2, 0)$, and ^{find} the unit vector in the direction in which the maximum value occurs.

Sol:-

$$\frac{\partial f}{\partial x} = f_x = 2xe^y, \quad f_x|_{(-2, 0)} = 2(-2)e^0 = -4$$

$$\frac{\partial f}{\partial y} = f_y = x^2 e^y, \quad f_y|_{(-2, 0)} = 4$$

$$\Delta f = f_x \hat{i} + f_y \hat{j} + f_z \hat{k}$$

$$= -4\hat{i} + 4\hat{j}$$

$$\hat{u} = \frac{\nabla f}{\|\nabla f\|}$$

$$\|\nabla f(-2, 0)\| = \sqrt{(-4)^2 + (4)^2} = \sqrt{32} = 4\sqrt{2}$$

$$\hat{u} = \frac{\nabla f(-2, 0)}{\|\nabla f(-2, 0)\|} = \frac{1}{4\sqrt{2}} (-4\hat{i} + 4\hat{j})$$

$$\hat{u} = -\frac{1}{\sqrt{2}} \hat{i} + \frac{1}{\sqrt{2}} \hat{j}$$

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Q 6

Find Duf at P

$$f(x, y, z) = ye^{xz} + z^2, \quad P(0, 2, 3)$$

$$v = \frac{2}{3} \hat{i} - \frac{3}{7} \hat{j} + \frac{6}{7} \hat{k}$$

Sol:-

$$\frac{\partial f}{\partial x} = f_x = ze^{xz}, \quad f_x|_{(0, 2, 3)} = 6$$

$$\frac{\partial f}{\partial y} = f_y = e^{xz}, \quad f_y|_{(0, 2, 3)} = 1$$

$$\frac{\partial f}{\partial z} = f_z = xy e^{xz} + 2z, \quad f_z|_{(0, 2, 3)} = 6$$

$$\Delta f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}$$

$$= f_x \hat{i} + f_y \hat{j} + f_z \hat{k}$$

$$\Delta f = 6\hat{i} + \hat{j} + 6\hat{k}$$

$$\hat{U} = \frac{\vec{U}}{|\vec{U}|} = \frac{6\hat{i} + \hat{j} + 6\hat{k}}{\sqrt{(6)^2 + (1)^2 + (6)^2}}$$

$$= \frac{6\hat{i} + \hat{j} + 6\hat{k}}{\sqrt{73}}$$

$$\Delta f = 6\hat{i} + \hat{j} + 6\hat{k}$$

$$\hat{U} = \frac{\vec{U}}{|\vec{U}|} = \frac{2\hat{i} - 3\hat{j} + 6\hat{k}}{\sqrt{(2)^2 + (-3)^2 + (6)^2}}$$

$$= \frac{2\hat{i}}{7} - \frac{3\hat{j}}{7} + \frac{6\hat{k}}{7}$$

$$\sqrt{\frac{4}{49} + \frac{9}{49} + \frac{36}{49}}$$

$$\hat{U} = \frac{2\hat{i}}{7} - \frac{3\hat{j}}{7} + \frac{6\hat{k}}{7}$$

$$D_u f = (6\hat{i} + \hat{j} + 6\hat{k}) \cdot \left(\frac{2}{7}\hat{i} - \frac{3}{7}\hat{j} + \frac{6}{7}\hat{k}\right)$$

$$= \frac{12}{7} - \frac{3}{7} + \frac{36}{7}$$

$$= \frac{45}{7}$$

⑦ $f(x, y, z) = \ln(x^2 + 2y^2 + 3z^2)$
 $P(-1, 2, 4)$

$$u = \frac{-3}{13}\hat{i} - \frac{4}{13}\hat{j} - \frac{12}{13}\hat{k}$$

Sol:-

$$\frac{\partial f}{\partial x} = f_x = \frac{1}{x^2 + 2y^2 + 3z^2} (2x), \quad f_x| = \frac{-2}{1+8+48}$$

$$2f_x = -2$$

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$$\frac{\partial f}{\partial y} = f_y = \frac{4y}{x^2 + 2y^2 + 3z^2}, \quad f_y| = \frac{8}{1+8+48}$$

$$= \frac{8}{57}$$

$$\frac{\partial f}{\partial z} = f_z = \frac{6z}{x^2 + 2y^2 + 3z^2}, \quad f_z| = \frac{24}{57}$$

$$\Delta f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}$$

$$= \frac{-2}{57} \hat{i} + \frac{8}{57} \hat{j} + \frac{24}{57} \hat{k}$$

$$\hat{U} = \frac{\vec{U}}{|\vec{U}|} = \frac{-\frac{3}{13} \hat{i} - \frac{4}{13} \hat{j} - \frac{12}{13} \hat{k}}{\sqrt{\left(-\frac{3}{13}\right)^2 + \left(-\frac{4}{13}\right)^2 + \left(-\frac{12}{13}\right)^2}}$$

$$= -\frac{3}{13} \hat{i} - \frac{4}{13} \hat{j} - \frac{12}{13} \hat{k}$$

$$D_{\vec{U}} f = \left(\frac{-2}{57} \hat{i} + \frac{8}{57} \hat{j} + \frac{24}{57} \hat{k} \right) \cdot \left(-\frac{3}{13} \hat{i} - \frac{4}{13} \hat{j} - \frac{12}{13} \hat{k} \right)$$

$$= \frac{6}{741} + \frac{32}{741} - \frac{288}{741}$$

$$= -\frac{250}{741}$$

13.6 Exa (1, 2, 3, 4)

Q6, 7, 8 Q17, Q18 Q45, Q46

Q8) $f(x, y, z) = \sin \pi y z$, $P(\frac{1}{2}, \frac{1}{3}, \pi)$

$$u = \frac{1}{\sqrt{3}} \hat{i} - \frac{1}{\sqrt{3}} \hat{j} + \frac{1}{\sqrt{3}} \hat{k}$$

Sol:-

$$f_x = yz \cos \pi y z, f_x \Big|_{\frac{1}{2}, \frac{1}{3}, \pi} = \frac{\pi}{3} \cos \frac{\pi}{6} = \frac{\sqrt{3}\pi}{6}$$

$$f_y = xz \cos \pi y z, f_y \Big|_{\frac{1}{2}, \frac{1}{3}, \pi} = \frac{\pi}{2} \cos \frac{\pi}{6}, \frac{\sqrt{3}\pi}{4}$$

$$f_z = xy \cos \pi y z, f_z \Big|_{\frac{1}{2}, \frac{1}{3}, \pi} = \frac{1}{6} \cos \frac{\pi}{6}, \frac{\sqrt{3}}{12}$$

$$\Delta f = \frac{\sqrt{3}\pi}{6} \hat{i} + \frac{\sqrt{3}\pi}{4} \hat{j} + \frac{\sqrt{3}}{12} \hat{k}$$

$$\hat{u} = \frac{\vec{u}}{|\vec{u}|} = \frac{\frac{1}{\sqrt{3}} \hat{i} - \frac{1}{\sqrt{3}} \hat{j} + \frac{1}{\sqrt{3}} \hat{k}}{\sqrt{\frac{1}{3} + \frac{1}{3} + \frac{1}{3}}} = \frac{\frac{1}{\sqrt{3}} \hat{i} - \frac{1}{\sqrt{3}} \hat{j} + \frac{1}{\sqrt{3}} \hat{k}}{1}$$

$$\hat{u} = \frac{1}{\sqrt{3}} \hat{i} - \frac{1}{\sqrt{3}} \hat{j} + \frac{1}{\sqrt{3}} \hat{k}$$

$$\frac{1}{\sqrt{3}} \hat{i} - \frac{1}{\sqrt{3}} \hat{j} + \frac{1}{\sqrt{3}} \hat{k}$$

$$Duf = \left(\frac{\sqrt{3} \pi}{6} \hat{i} + \frac{\sqrt{3} \pi}{4} \hat{j} + \frac{\sqrt{3}}{12} \hat{k} \right)$$

$$\left(\frac{1}{\sqrt{3}} \hat{i} - \frac{1}{\sqrt{3}} \hat{j} + \frac{1}{\sqrt{3}} \hat{k} \right)$$

$$Duf = \frac{1\pi}{6} - \frac{1\pi}{9} + \frac{1}{12} = \frac{1-\pi}{12}$$

13.6

Q17. $f(x, y, z) = \frac{z-x}{z+y}$; $P(1, 0, -3)$

; $a = -6\hat{i} + 3\hat{j} - 2\hat{k}$

Find directional derivative of f at P in the direction of a .

Sol: -

$$\frac{\partial f}{\partial x} = f_x = \frac{-1}{z+y}, \quad f_x|_{(1, 0, -3)} = \frac{-1}{-3+0}$$

$$\frac{\partial f}{\partial y} = \frac{(z+y)^{-1}}{(z+y)^2} = \frac{-1}{(z+y)^2} = \frac{1}{3}$$

$$\frac{\partial f}{\partial y} = f_y = \frac{(z+y)(0) - (z-x)(1)}{(z+y)^2}$$

$$f_y = -\frac{(z-x)}{(z+y)^2}$$

$$\frac{\partial f}{\partial z} = f_z = \frac{(z+y)^2(1) - (z-x)(1)}{(z+y)^2}$$

$$= \frac{z+y - z+x}{(z+y)^2} = \frac{x+y}{(z+y)^2}$$

$$f_y|(1, 0, -3) = \frac{-(-3-1)}{(-3+0)^2} = \frac{4}{9}$$

$$f_z|(1, 0, -3) = \frac{1+0}{(-3+0)^2} = \frac{1}{9}$$

$$\Delta f = f_x \hat{i} + f_y \hat{j} + f_z \hat{k}$$

$$= \frac{1}{3} \hat{i} + \frac{4}{9} \hat{j} + \frac{1}{9} \hat{k}$$

$$\hat{u} = \frac{\vec{a}}{|\vec{a}|} = \frac{-6\hat{i} + 3\hat{j} - 2\hat{k}}{\sqrt{(-6)^2 + (3)^2 + (-2)^2}}$$

$$= \frac{-6\hat{i} + 3\hat{j} - 2\hat{k}}{\sqrt{36+9+4}} = \frac{-6\hat{i} + 3\hat{j} - 2\hat{k}}{7}$$

$$\therefore D_{\vec{u}} f = \Delta f \cdot \hat{u}$$

$$= \left(\frac{1}{3} \hat{i} + \frac{4}{9} \hat{j} + \frac{1}{9} \hat{k} \right) \cdot \left(\frac{-6\hat{i} + 3\hat{j} - 2\hat{k}}{7} \right)$$

$$= -\frac{6}{21} \hat{i} + \frac{12}{63} \hat{j} - \frac{2}{63} \hat{k}$$

$$= -\frac{2}{7} \hat{i} + \frac{4}{63} \hat{j} - \frac{2}{63} \hat{k}$$

$$= \frac{-18}{63} + \frac{12}{63} - \frac{2}{63}$$

$$= \frac{-8}{63}$$

18) $f(x, y, z) = e^{x+y+3z}$; $P(-2, 2, -1)$; $\nabla f = 20\hat{i} - 4\hat{j} + 5\hat{k}$

Sol:-

$$\frac{\partial f}{\partial x} = f_x = e^{x+y+3z} \frac{\partial}{\partial x} (x+y+3z)$$

$$\frac{\partial f}{\partial x} = f_x = e^{x+y+3z}$$

$$\frac{\partial f}{\partial y} = f_y = e^{x+y+3z} \frac{\partial}{\partial y} (x+y+3z)$$

$$\frac{\partial f}{\partial y} = f_y = e^{x+y+3z}$$

$$\frac{\partial f}{\partial z} = f_z = e^{x+y+3z} \frac{\partial}{\partial z} (x+y+3z)$$

$$\frac{\partial f}{\partial z} = f_z = 3e^{x+y+3z}$$

$$\frac{\partial f}{\partial x} = f_x|_{(-2, 2, -1)} = e^{-2+2+3(-1)} = e^{-3}$$

$$= f_y|_{(-2, 2, -1)} = e^{-2+2+3(-1)} = e^{-3}$$

~~Do~~

$$f_z|(-2, 2, -1) = 3e^{-2+2+3(-1)} = 3e^{-3}$$

$$\Delta f = f_x \hat{i} + f_y \hat{j} + f_z \hat{k}$$

$$= e^{-3} \hat{i} + e^{-3} \hat{j} + 3e^{-3} \hat{k}$$

$$\Delta f = e^{-3} (\hat{i} + \hat{j} + 3\hat{k})$$

$$a = \frac{\vec{a}}{|\vec{a}|} = \frac{20\hat{i} - 4\hat{j} + 5\hat{k}}{\sqrt{(20)^2 + (-4)^2 + (5)^2}}$$

$$= \frac{20\hat{i} - 4\hat{j} + 5\hat{k}}{21}$$

$$D_{\vec{a}} f = \Delta f \cdot \hat{a}$$

$$= e^{-3} (\hat{i} + \hat{j} + 3\hat{k}) \cdot \frac{1}{21} (20\hat{i} - 4\hat{j} + 5\hat{k})$$

$$= \frac{e^{-3} (20 - 4 + 15)}{21}$$

$$= \frac{31}{21} e^{-3}$$

~~Q45~~ Find the gradient of f at the indicated point.

Q45

$$f(x, y, z) = y \ln(x+y+z); (-3, 4, 0)$$

Sol:-

$$\frac{\partial f}{\partial x} = f_x = \frac{y}{x+y+z} \frac{\partial}{\partial x} (x+y+z)$$

$$= \frac{y}{x+y+z}$$

$$\frac{\partial f}{\partial y} = f_y \ln(x+y+z) + y \left(\frac{1}{x+y+z} \right) \frac{\partial}{\partial y} (x+y+z)$$

$$= \ln(x+y+z) + \frac{y}{x+y+z}$$

$$\frac{\partial f}{\partial z} = f_z = \frac{y}{x+y+z}$$

$$f_x | (-3, 4, 0) = \frac{4}{-3+4+0} = +\frac{4}{1} = 4$$

$$f_y|(-3, 4, 0) = \ln(-3+4+0) + \left(\frac{-4}{1}\right)$$

$$= \ln(1) - \frac{4}{1} = 0 - 4 = -4$$

$$f_z|(-3, 4, 0) = +\frac{4}{1} = 4$$

$$\Delta f = f_x \hat{i} + f_y \hat{j} + f_z \hat{k}$$

$$= +\frac{4}{1} \hat{i} + \frac{-4}{1} \hat{j} + \frac{4}{1} \hat{k}$$

$$= 4\hat{i} - 4\hat{j} + 4\hat{k}$$

$$46) f(x, y, z) = y^2 z \tan^3 x; \left(\frac{\pi}{4}, -3, 1\right)$$

Sol:-

$$\begin{aligned} \frac{\partial f}{\partial x} = f_x &= y^2 z \cdot 3 \tan^2 x \sec^2 x \\ &= 3y^2 z \tan^2 x \sec^2 x \end{aligned}$$

$$\frac{\partial f}{\partial y} = f_y = 2yz \tan^3 x$$

$$\frac{\partial f}{\partial z} = f_z = y^2 \tan^3 x$$

$$f_x \left| \left(\frac{\pi}{4}, -3, 1 \right) \right. = 3(-3)^2 (1) \tan^2 \left(\frac{\pi}{4} \right) \sec^2 \left(\frac{\pi}{4} \right)$$

$$\because \tan \left(\frac{\pi}{4} \right) = 1 \text{ \& } \sec \left(\frac{\pi}{4} \right) = \sqrt{2}$$

$$\begin{aligned} f_x \left| \left(\frac{\pi}{4}, -3, 1 \right) \right. &= 3(-3)^2 (1) (1)^2 \cdot (\sqrt{2})^2 \\ &= 3 \times 9 \times 1 \times 2 \\ &= 54 \end{aligned}$$

$$\begin{aligned} f_y \left| \left(\frac{\pi}{4}, -3, 1 \right) \right. &= 2(-3)(1)(1) \\ &= -6 \end{aligned}$$

$$\begin{aligned} f_z \left| \left(\frac{\pi}{4}, -3, 1 \right) \right. &= (-3)^2 (1) \\ &= 9 \end{aligned}$$

$$\Delta f_z \hat{i} + f_y \hat{j} + f_z \hat{k}$$

$$= 54 \hat{i} + (-6 \hat{j}) + 9 \hat{k}$$

$$= 54 \hat{i} - 6 \hat{j} + 9 \hat{k}$$